6 **(a)** A sample of water is contaminated with radioactive iodine-131 ($^{131}_{53}I$).
The activity of the iodine-131 in 1.0 kg of this water is 460 Bq.
The half-life of iodine-131 is 8.1 days.

(i) Calculate the number of iodine-131 atoms in 1.0 kg of this water.

number = …….. [2]

(ii) An amount of 1.0 mol of non-contaminated water has a mass of 18 g.

Calculate the ratio

$$\frac{\text{number of molecules of water in 1.0 kg of contaminated water}}{\text{number of atoms of iodine-131 in 1.0 kg of contaminated water}}$$

ratio = ……. [2]

(iii) An acceptable limit for the activity of iodine-131 in water has been set as 170 Bq kg^{-1}.

Calculate the time, in days, for the activity of the contaminated water to be reduced to this acceptable level.

time = …… days [2]

(b) A milk sample is to be tested for evidence of radioactive contamination with the radioactive nuclide strontium-90 (^{90}Sr), using a Geiger-Muller tube. The two stages involved in the decay of ^{90}Sr are described by the following two equations.

Stage 1: $^{90}_{38}Sr \rightarrow ^{90}_{39}Y + \beta$

Stage 2: $^{90}_{39}Y \rightarrow ^{90}_{40}Zr + \beta$

At each stage, a β-particle is emitted. At the end of stage 2, the product zirconium-90 (^{90}Zr) is stable.

(i) For stage 1, the emitted β-particles have a range of energies up to a maximum of 0.55 MeV. Use conservation laws to explain why this range of energies leads to the suggestion that another particle is emitted by the decaying strontium-90 nucleus together with the β-particle.

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…… [4]

(ii) Explain why it is difficult to measure the half-life of the beta decay of the strontium-90 experimentally.

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…… [1]